

What makes a strong project in 2026?

LLMs made implementation cheap. They did not make good questions cheap.

What we value

Rewarded

- ✓ A clear research question you can answer in ten weeks
- ✓ Baselines, ablations, several seeds — evidence, not one number
- ✓ Honest error analysis: when does it fail, and why?
- ✓ A negative result with an explanation
- ✓ Code someone else can run

Not enough

- ✗ “We applied model X to dataset Y and got 93%”
- ✗ No baseline, no ablation, one seed
- ✗ Results you cannot explain when asked
- ✗ Reproducing a paper that already has public code
- ✗ A repository written the night before the deadline

Novelty is not required. Depth, clarity and critical thinking are.

Four kinds of projects — combinations welcome

1. Reproduce and stress-test

A paper without public code — or a reproducibility study: does the claim survive other seeds, datasets, ablations, distribution shift?

2. New domain or task

Apply models to a genuinely new problem. You must argue why it is hard and which question you answer — not just report accuracy.

3. Method or theory

A new architecture, layer, objective, or training procedure; A property of a model you can prove. Combining two existing methods is fine.

4. Evaluation and analysis

A benchmark, a failure-mode study, model internals, uncertainty, or the interpretability of an existing model.

Ambitious means a sharp question and a rigorous evaluation — not the biggest model you can find.

Calibration: projects from a previous offering

(2024 — courtesy of my colleague Hao)

- Disentangled representation learning for causal VAEs
- Train- and test-time domain generalization
- Privacy-preserving synthesis of high-dimensional data
- Image denoising under ground-truth uncertainty
- Finding interpretable factors in generative models
- Toward robust image classification
- Spatial attention for medical image segmentation
- Joint reconstruction and segmentation of the left atrium from 3D LGE-MRI
- Graph attention autoencoders
- Learning to explain your plan: GNN-augmented Dyna
- Semi-supervised named entity recognition
- Music generation with GANs and Transformers

Directions you can consider (but not limited to)

1. **Model Merging & Weight Arithmetic:** Combining two specialized models into one by mathematically blending their weights, without retraining from scratch.
2. **Generative Modeling (Diffusion):** Teaching networks to generate new, high-fidelity images, audio, or textures by removing noise step by step.
3. **Time Series & Sequential Signals:** Predicting future trends and detecting anomalies in continuous sensor readings, medical telemetry, and real-world event streams.
4. **AI for Science & Physics:** Embedding physical laws (conservation of mass, energy, fluid dynamics) directly into the neural network's loss function to simulate natural phenomena.
5. **Geometric DL & Graph Networks:** Running neural networks on relational, non-grid data like molecular structures, chemical bonds, and transport networks.
6. **Multimodal Learning:** Connecting different sensory signals inside a shared latent space, like linking audio waveforms to video frames to locate sound sources.
7. **Trustworthy & Safe AI:** Measuring model uncertainty with statistical guarantees, stopping models from learning spurious background shortcuts or memorizing private data.

These are fields description not project names

Pre-LLMs, “CNNs for X diagnosis” was an interesting project.

**In 2026 it needs a question:
shortcuts? robustness? calibration? uncertainty? interpretability? benchmarking?**

If you would like feedback on the project topic

Bring one paper you read and one or few questions — after class or by email.

My work is in trustworthiness, safety, interpretability and model uncertainty and I am interested broadly in those and related topics.

Compute and data: plan for them in the proposal

- **Compute:** Google Colab (free tier is limited and sessions time out) and iLab machines at Rutgers. Benchmark one training run in week one.
- **Ambitious $\neq$ compute-heavy:** small models plus careful experiments beat one giant run. Pretrained models and fine-tuning are your friends.
- **Data access:** public benchmarks are immediate; credentialed data (e.g. MIMIC) can take weeks of paperwork — start now or choose otherwise.
- **Privacy:** never upload confidential, private, or access-restricted data or code to external AI tools.
- **Reproducibility:** a repository from day one; fixed seeds; a README that actually runs.

Questions I get often

Ask the ones I missed — after class or on Canvas.

Can the project be on my own research?

Yes — if it is related to this class and the work is genuinely new.

- The question should be about deep learning as we study it here.
- The semester's work must be new: not a write-up of results you already have, and not work submitted for another course.
- **In the proposal**, state what existed before September and what you will do now.
- If your advisor is involved, make sure they are on board, and say who does what.

Can I do an independent study with you?

I do independent studies with students I have already worked with in some capacity.

- **This project is that path:** do a strong project this semester.
- If it goes well and we both want to continue, we can set up an independent study or thesis supervision to continue it next semester.

Can my results be published?

Maybe — but it is not the goal of the course.

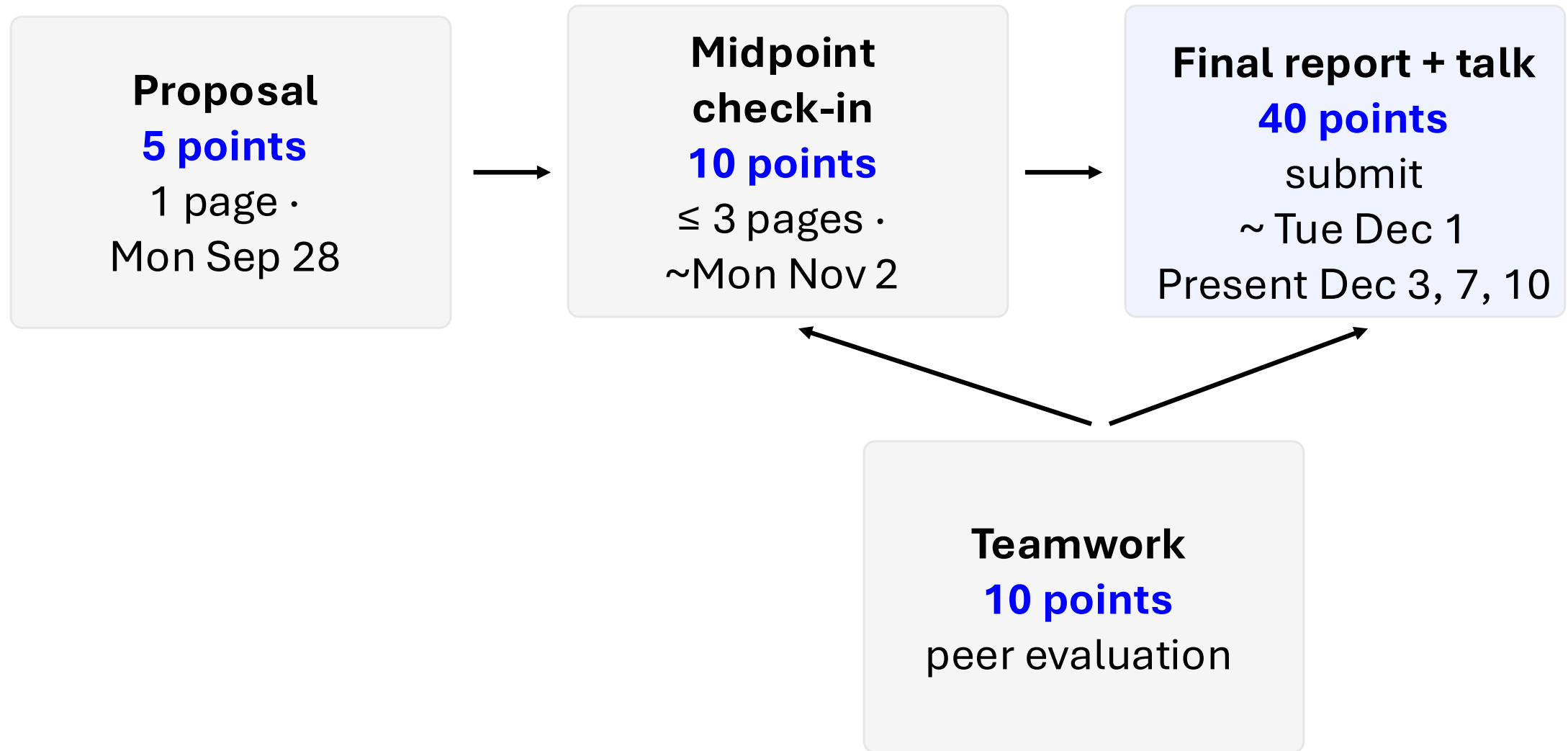
- Most course projects are not publication-ready in December, and the grade does not depend on it.
- **If publication is your goal**, tell me early and aim high: I will give feedback as you choose the topic and as results come in. We can keep chatting throughout the semester. Alternatively, if you worked on the project and your results are interesting, you can think about publication after the fact in December. Remember, research does not always work the way you hoped for in the beginning.
- **What I can promise:** honest feedback, and if the results are strong, help framing them for a workshop or a conference.
- **What I cannot promise:** a publication. It is your project and your work — authorship follows contribution.

Can I get extra help on the project from outside of the class?

Yes — talk to me.

- I try to be as inclusive as possible: pointing you to a graduate student, a dataset, compute, or a collaborator.
- **Two conditions:** the project remains your team's own work, and everyone and everything that contributed is acknowledged in the report — people, tools, prior code, LLMs.
- **Ask early:** the week before a deadline is too late to add support.

How the project is graded



Milestone 1: proposal

- **Research question:** one sentence — what would a positive answer look like? A negative one?
- **Motivation and prior work:** 3–5 papers you have actually read; what your study adds
- **Data, compute, method:** what you will use — and whether you have access to it today
- **Evaluation plan:** metrics, baselines, ablations; how you avoid data leakage
- **Risks and fallback:** the most likely way this fails, and what you do then
- **Plan and roles:** week-by-week milestones; who does what; anticipated LLM use

Milestone 2: midpoint check-in

- **Report:**

initial results, lessons learned,
successes and challenges;
did you change direction? what is next?

- **Repository:** link to your GitHub

A negative result at the midpoint is fine. No baseline at the midpoint is not.

Milestone 3: final report and presentation (40 points)

- **Every team, same deadline**
- **Report:** 5-8 pages, NeurIPS-style template, unlimited appendix; link to your GitHub
- **Content:** motivation, methods, main results, experiments, observations, lessons learned

Present: Dec 3(?), 7 and 10

- **~12-15 min talk + 3 min Q&A**
(final length once teams are known)
- **Order randomized**, announced later
- **Every team member** presents part of the talk and answers questions
- **Audience:** feedback and questions on each talk counts toward engagement

Same deadline for everyone — your presentation slot does not change it.

Teams

- **Size:** 1–3 people. A team of 4 needs an ambitious scope to justify it; 5 or more is discouraged.
- **Declare by Sep 28:** team and tentative topic in the team declaration form (link on Canvas and in the syllabus)
- **Changes of team or topic:** free until the midpoint; afterwards only with approval
- **Roles:** the commit history should show everyone's work

Working alone is allowed — but two people who argue about the evaluation usually go further.

Let's move!

Let's form a line based on the phone (pr laptop) battery right now.

0% near the podium, 100% at the right end

You'll have to ask each other

<https://forms.gle/RcVMXJgyRCPAFrYp8>

Break-out chat (groups of 3–4, 7 minutes)

Go to google form

- 1. Discuss:** your two directions and the LLM's strongest objection to each.
- 2. Score direction together (examples of questions):**
 1. Is it a question or a task?
 2. What do we learn if the answer is no?
 3. Could we have a first result by Oct 12?
 4. Are the data and compute available today?
- 3. Pick either:** the group's best question or the one that *looked* best but is not. Post in the form.

